

Medical Robotics

Comprehensive Study Notes for Exam Preparation

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Chapter 1

Introduction and Classification

1.1 Definition of a Medical Robot

The definition of a robot in the medical field is multifaceted. According to Michael Brady (1980), robotics is "*the intelligent connection between perception and action*". This definition perfectly fits medical robotics, where sensors (perception) read physical interactions and anatomy, and actuators (action) execute commands mediated by the surgeon's intelligence or a control algorithm. From a regulatory standpoint, the IEC TR 60601-4-1 directive defines a medical robot as "medical electrical equipment with a degree of autonomy intended to perform tasks." The paradox of the surgical robot is well summarized by J. Moser's quote: "*Just a Million Dollar Needle Holder*." The true engineering challenge is proving that this highly sophisticated tool objectively improves clinical outcomes compared to traditional manual surgery.

1.2 The Evolution of Surgery and its Tasks

Surgical evolution is driven by the need to resolve a fundamental trade-off: maximizing the therapeutic effect while minimizing collateral damage (patient trauma).

- **Open Surgery:** Guarantees maximum dexterity and 3D vision for the surgeon but causes large incisions, high risk of infections, and prolonged hospital stays.
- **Minimally Invasive Surgery (MIS - Laparoscopy):** Operations are performed through 5-12 mm incisions using trocars. Trauma drops significantly, but the surgeon faces the *fulcrum effect* (inverted hand-tool motion), loss of depth perception (2D vision), the need for a camera assistant, and poor ergonomics.
- **Micro/Nano Surgery:** Therapies at the cellular level. The physical obstacle here is that at microscopic scales (e.g., 100 μm blood vessels), the surgeon's physiological hand tremor (approx. 182 μm) is larger than the target itself.

Medical robotics, specifically with the *da Vinci* system (introduced in 1999), emerged to merge the benefits of both paradigms: restoring the dexterity of Open Surgery (via 7-DOF articulated EndoWrist instruments) while preserving the low invasiveness of MIS.

1.3 Classification Schemes

Surgical robotic systems are complex and multifaceted. Several classification schemes are presented in the literature, as each highlights a different engineering and clinical aspect.

1.3.1 Troccaz Classification (Interaction Modality, 2001)

Based on the level of physical interaction between the machine and the human:

- **Passive Systems:** Unactuated, manually driven by the surgeon. The system acts as a spatial navigator. Example: *Viewing Wand* (an articulated arm with encoders that displays the scalpel tip's position on a pre-operative CT scan).
- **Active Systems:** Execute a surgical plan autonomously under human supervision. Example: *ROBODOC* for autonomous orthopedic bone milling; *CyberKnife* for frameless stereotactic radiosurgery.
- **Interactive Systems (Shared Control):** The machine and the human share physical control of the instrument. They are divided into:
 - *Semi-active:* The robot autonomously positions a mechanical guide and then locks its motors. The surgeon inserts the tool through the guide (e.g., NeuroMate).
 - *Synergistic:* The constraint is "virtual" and programmed via software, such as Virtual Fixtures (e.g., PADyC, Acrobot/MAKO).
- **Teleoperated Systems:** The surgeon is physically decoupled from the patient and operates a master console. A slave robot replicates the movements, applying tremor filtering and *motion scaling* (e.g., da Vinci).

1.3.2 Other Classifications: Taylor, Joskowicz, Dario

- **Taylor (2003):** Introduces the concept of *Computer Integrated Surgery (CIS)*. Distinguishes between CAD/CAM systems (where image registration plays a vital role, e.g., ROBODOC) and *Surgical Assistants* (e.g., *AESOP*, a voice-controlled arm to hold the endoscope, eliminating human assistant fatigue).
- **Joskowicz (2005):** Classifies based on the type of support, ranging from patient-specific passive jigs to intraoperative navigation, floor-mounted robots, and finally **patient-mounted robots** (e.g., *Mazor/Renaissance* for spinal surgery, which are fixed to the patient's bone, making the system immune to respiratory movements and eliminating the need for real-time tracking).
- **Paolo Dario (2004):** Creates a 2D matrix crossing the *Type of Access* (Traditional, MIS, Endoluminal) with the *Type of Interaction*.

1.3.3 Levels of Autonomy (LoA) and Regulations

Adapted from the SAE autonomous driving scale, the LoA framework ranges from 0 (no autonomy) to 5 (full autonomy, currently non-existent in clinical practice). The standard *da Vinci* is **LoA 1** (the surgeon continuously controls everything; the robot only filters tremor). A robot like *ROBODOC* is **LoA 2** (task-level autonomy: the human initiates the process, and the robot executes the milling). From a regulatory perspective, the **IEC 80601-2-77** standard mandates the definition and demonstration of "essential performance" based on the device's level of invasiveness.

Chapter 2

Kinematic Design and the RCM Constraint

Kinematic design is strictly driven by the surgical task. Engineers do not design "universal" robots, but rather optimize workspace, dexterity, and safety around the specific operation to be performed.

2.1 Serial Architectures: SCARA vs. Anthropomorphic

- **SCARA (e.g., ROBODOC):** The architecture of choice for milling rigid surfaces. Its workspace is an annular cylinder that perfectly "wraps" around a patient lying on the operating table without wasting volume. The primary axis is prismatic and vertical: this provides **intrinsic gravitational safety** (if power fails, the arm cannot fall sideways onto the patient). Arm singularities are known and confined to the outer boundaries of the workspace.
- **Anthropomorphic (e.g., CyberKnife):** Features a spherical workspace. It allows approaches from multiple angles, which is fundamental for non-isocentric radiosurgery. However, in the event of power loss, the cantilevered links would collapse onto the patient due to gravity, necessitating heavy counterweights or highly reliable redundant braking systems.

2.2 The Remote Center of Motion (RCM) Constraint

In MIS, the instrument shaft passes through the abdominal wall at a fixed point (the trocar). No transverse forces can be generated at this point, otherwise the tissue would tear. Kinematically, this constraint forces the robot to move such that a stationary point (the RCM) exists along the instrument shaft, around which it can only rotate and translate axially (providing 4 DOFs inside the abdomen).

There are four engineering methods to satisfy the RCM constraint:

1. **Passive Joints (e.g., Zeus system):** The robot's distal joints, near the trocar, are free (unactuated). When the surgeon moves the system, the resistance of the abdominal wall forces the mechanics to align naturally. It is intrinsically safe but lacks precision due to mechanical friction.
2. **Mechanical RCM (e.g., da Vinci Patient Cart):** The structure is geometrically designed (e.g., using an articulated parallelogram) so that, regardless of motor angles, the

instrument axis always passes through a fixed focal point in space. The advantage is that the RCM is hardware-guaranteed; the disadvantage is the massive mechanical footprint above the patient and the need to physically align this fixed point exactly with the incision.

3. **Kinematic Control (Software RCM):** The robot is a standard, redundant serial arm (e.g., 6 or 7 DOFs). The RCM is mathematically enforced via software as a zero-velocity constraint in Cartesian space: $\dot{p}_{RCM} = J_{RCM}(q)\dot{q} = 0$. The motors perform *null-space* motions to move the tip while keeping the trocar point stationary. This allows for compact, multi-purpose robots, but patient safety relies entirely on software reliability.
4. **Force Control:** A force/torque sensor is installed on the wrist. If the shaft presses against the trocar, the sensor detects it and commands the motors to yield to the push, minimizing the transverse force. Very flexible, but sensors introduce complexity, cost, and sterilization issues.

2.3 Degrees of Freedom Analysis (Grübler's Formula)

To determine how many joints the external robotic arm needs, Grübler's formula is used. An arm inserted into an abdomen closes a kinematic chain with the operating table, where the trocar acts as a 4-DOF virtual joint. The formula for a closed loop (where the number of joints n equals the number of links l) simplifies to:

$$m = \sum d_i - 6(n + 1 - l) \implies m = \text{Total DOFs} - 6 \quad (2.1)$$

Let x be the number of actuated DOFs of the external arm. The total DOFs are $x + 4$ (the 4 DOFs of the trocar). If the surgeon requires an internal mobility $m = 4$ (e.g., targeting a needle tip), we have: $4 = x + 4 - 6 \implies x = 6$. Therefore, 6 external DOFs are required to position the tool tip internally while respecting the trocar constraint.

Chapter 3

Parallel Architectures and Special Applications

Although serial manipulators are the majority, **parallel** architectures (where multiple independent chains connect the base to the end-effector) are indispensable for tasks requiring lightning-fast dynamics, high stiffness, and a high payload-to-weight ratio.

3.1 Serial vs. Parallel Manipulators

In **parallel manipulators**, joint errors average out rather than accumulating. Since heavy motors are often fixed to the base, the inertia of the moving parts is extremely low, allowing for very high operating frequencies. Mathematically, the **Inverse Kinematics** of parallel robots is generally decoupled and solvable in closed analytical form: given the platform's pose, each leg becomes a simple, independent serial problem. Conversely, **Direct (Forward) Kinematics** is extremely complex. Even for a simple planar 3-RPR mechanism, the solution leads to a 6th-degree polynomial, often requiring numerical iterative methods. The workspace of a parallel robot is very restricted (the intersection of the individual legs' workspaces). While a flaw in industry, this becomes a **feature of intrinsic safety** in surgery: in the event of a controller runaway, the hardware physically cannot exceed a confined volume.

3.2 Singularity Analysis

The differential kinematics of a parallel manipulator is governed by two Jacobian matrices, derived by differentiating the closure constraint $\mathbf{H}(\mathbf{X}, \boldsymbol{\theta}) = 0$:

$$\mathbf{U}\dot{\mathbf{X}} + \mathbf{V}\dot{\boldsymbol{\theta}} = \mathbf{0} \quad (3.1)$$

Because there are two Jacobians, three distinct types of singularities emerge:

1. **Type 1 (Serial Singularities):** $\mathbf{V}$ loses rank ($\det(\mathbf{V}) = 0$). These occur at the boundary of the workspace when legs are fully extended or folded. Non-zero joint velocities ($\dot{\boldsymbol{\theta}} \neq 0$) exist that produce no end-effector velocity ($\dot{\mathbf{X}} = 0$).
2. **Type 2 (Parallel/Local Singularities):** $\mathbf{U}$ loses rank ($\det(\mathbf{U}) = 0$). This is the most critical case. A non-zero end-effector velocity $\dot{\mathbf{X}} \neq 0$ exists **even when all active motors are physically locked** ($\dot{\boldsymbol{\theta}} = 0$). The mechanism gains an uncontrollable infinitesimal degree of freedom and cannot resist external forces. These occur *inside* the workspace and **must be strictly avoided in surgery**, as they lead to loss of tool control.

3. **Type 3 Singularities:** Both matrices lose rank simultaneously, due to degenerate architectural choices (e.g., exact symmetric proportions between base and moving platform).

3.3 Case Study 1: The Cardiolock (Exploiting Singularities)

The **Cardiolock** is an active stabilizer designed for beating-heart coronary surgery. Unlike passive stabilizers that leave millimeter-scale residual motions, the Cardiolock actively moves to cancel myocardial motion, keeping the surgical area virtually motionless.

Compensation requires a very high bandwidth (heart frequencies $\sim 10\text{Hz}$), forcing designers to use **piezoelectric actuators** (zero friction, zero backlash). The flaw of piezo actuators is their **extremely limited stroke**, on the order of $100\text{-}200\ \mu\text{m}$. How can one compensate for millimeters of heartbeat with micrometers of stroke? The brilliant engineering intuition behind Cardiolock was to employ a planar 3-PRR mechanism and deliberately operate it ****in close proximity to a Type 2 parallel singularity****. Kinematics dictates that near a Type 2 singularity, the ratio of end-effector velocity to actuator velocity approaches infinity. By exploiting this phenomenon, a microscopic translation of the piezo actuator is kinematically "amplified" into a vast rotation of the robot's wrist, solving the stroke problem without compromising bandwidth.

3.4 Case Study 2: Pentagon Mechanism (In-Vivo Microscopy)

Similarly to Cardiolock, stabilizing a microscope image on living animals requires compensating for large displacements at high frequencies. A 5-bar closed-loop parallel mechanism ("Pentagon") is used, pushed at the base by two piezoelectric actuators. Again, the goal is to optimize the link geometry (by analyzing the Jacobian and its eigenvalues) to maximize the motion amplification factor isotropically, allowing the microscope to track the target cell without losing focus.

Chapter 4

Interaction Control: Impedance and Admittance

When a medical robot comes into contact with the patient or the surgeon, pure position control (which forces the robot to reach fixed coordinates regardless of the environment) is dangerous. If a position-controlled robot hits an unexpected bone, it will increase motor current to its limit to reduce the position error, destroying tissues. It is mandatory to control the dynamic relationship between motion and exchanged force. This is achieved via **Impedance** and **Admittance** control.

Assuming a low-level controller is already compensating for gravity and Coriolis forces (feedback linearization in Cartesian space), the machine's dynamics can be reduced to a controllable mass m :

$$m\ddot{p} = u + F_{ext} \quad (4.1)$$

where u is the control command and F_{ext} is the external force.

4.1 Impedance Control

In impedance control, **the input to the model is a kinematic perturbation (position/velocity) and the output is a commanded force**. The goal is to force the robot to follow this desired target equation:

$$m_d(\ddot{p} - \ddot{p}_d) + b(\dot{p} - \dot{p}_d) + k(p - p_d) = F_{ext} \quad (4.2)$$

where m_d, b, k are the desired apparent mass, damping, and stiffness, respectively, and p_d is the nominal trajectory. Solving for the desired acceleration $\ddot{p}$ and substituting it into the robot's dynamics yields the control law:

$$u = m\ddot{p}_d + \frac{m}{m_d} \left(b(\dot{p}_d - \dot{p}) + k(p_d - p) \right) + \left(\frac{m}{m_d} - 1 \right) F_{ext} \quad (4.3)$$

Crucial Detail: If the designer chooses **not** to alter the robot's perceived inertia, setting $m_d = m$ (apparent mass equals real mass), the final term containing F_{ext} cancels out completely. This means that **impedance control (without inertial reshaping) requires no force/torque sensor at the end-effector**. It simply becomes a Proportional-Derivative (PD) controller with acceleration feedforward. This scheme perfectly suits robots with low-inertia direct-drive actuators and high-efficiency transmissions (e.g., cable-driven), where *torque control* can be commanded directly to the motors in a *backdrivable* manner.

4.2 Admittance Control

Admittance control reverses the causality: **the input is force, and the output is a motion correction**. Here, **a force sensor is mandatory**. The surgeon or environment applies a force F_{ext} , which is read by the sensor and passed to a virtual dynamic model to calculate a target position modification Δp_r :

$$\Delta p_r = \frac{F_{ext}}{m_d s^2 + b s + k} \quad (4.4)$$

The new reference $p_r = p_d + \Delta p_r$ is then fed into a highly stiff, high-gain inner PID position controller, which forces the motors to track p_r .

This approach is fundamental for retrofitting conventional, heavy industrial robots. Industrial robots often use Harmonic Drives, which generate massive friction and make the arm **non-backdrivable** (it cannot be pushed physically). With admittance control, the force sensor detects the push and the software "fakes" compliance by continuously updating position targets, masking the arm's intrinsic mechanical rigidity.

Chapter 5

Virtual Fixtures and Synergistic Systems

5.1 What are Virtual Fixtures?

Introduced in the 1990s by L.B. Rosenberg (US Air Force), **Virtual Fixtures (VF)** are "*abstract perceptual overlays*" introduced into the control interface. They are not hardware obstacles, but computational limits that simulate the effect of mechanical guides (like a ruler), vastly increasing the safety and precision of shared-control and teleoperated tasks.

- **Guidance VFs (GVF)**: Actively attract or channel the robot's motion along a preferred 1D path or 2D surface (e.g., a straight needle insertion).
- **Forbidden-Region VFs (FRVF)**: Prevent access to delicate anatomical zones, acting as virtual walls.

These fixtures can be modeled as **Hard** (absolute prohibition to deviate, the robot fully absorbs non-compliant forces) or **Soft** (provides clear elastic tactile resistance, but a sufficient force from the surgeon can override the constraint, providing an exit strategy if pre-operative planning was flawed).

5.2 Mathematical Modeling of GVFs

To mathematically construct an admittance-based GVF, one defines a matrix $\mathbf{P}(t)$ representing the space of desired motion directions point by point. The orthogonal projection operator onto this subspace is:

$$\Pi_P = \mathbf{P}(\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \quad (5.1)$$

When the operator applies a force f , the control law filters the directions:

$$v = c \left(\underbrace{\Pi_P f}_{\text{useful force}} + \lambda_{\perp} \underbrace{(\mathbf{I} - \Pi_P) f}_{\text{transverse force}} \right) \quad (5.2)$$

where c is the nominal admittance and $\lambda_{\perp} \in [0, 1]$ is the attenuation factor. If $\lambda_{\perp} = 0$, we have a Hard GVF (transverse forces are multiplied by zero). If $\lambda_{\perp} > 0$, we obtain a Soft GVF.

5.3 Historical Synergistic Systems: PADyC and Acrobot

Synergistic systems represent the physical implementation of VFs in shared-control tasks.

5.3.1 PADyC (Passive Arm with Dynamic Constraints)

Developed by TIMC Grenoble, PADyC is a triumph of passive mechatronics. It uses no motors to push the tool. Instead, each joint features two opposed *freewheels* connected to motors. A freewheel blocks motion in one direction but allows it in the other. By regulating the speed of the internal motors, PADyC defines a "window of admissible velocities" $[\omega_{min}, \omega_{max}]$ for each joint. The system is **intrinsically safe**, but the underlying mathematical problem is daunting: translating Cartesian safety constraints into a Window of Admissible Configurations (WAC) for each joint in configuration space is computationally extremely expensive, limiting its clinical success.

5.3.2 Acrobot / MAKO (Active Constraints Robot)

Developed at Imperial College (now MAKO by Stryker) for knee arthroplasty. Unlike PADyC, it uses an active arm and **impedance control** to generate a FRVF. The intervention area is divided into three regions based on the distance d from the safe bone boundary:

1. **Region I (Safe Zone, $d > D1$):** The robot has very low impedance (*free motion*). The surgeon moves transparently.
2. **Region II (Transition Zone, $0 < d < D1$):** As the tool approaches the boundary, the robot's impedance increases progressively in the normal direction, becoming stiffer, while remaining compliant tangentially.
3. **Region III (Forbidden Zone, $d < 0$):** Beyond the boundary, PD gains max out; the robot brakes sharply, preventing further penetration.

The gradual stiffness increase in RII is crucial: a sudden step-function increase in stiffness would trigger unstable vibrations due to the violent force transient on the motors.

Chapter 6

Robotic Registration

Registration is the mathematical heart of Computer Integrated Surgery. It corresponds to finding the spatial transformation (Rotation $\mathbf{R}$ and Translation $\mathbf{t}$) that maps the pre-operative digital coordinates (CT scans, surgical plans) to the physical space in the OR where the robot moves. It must be distinguished from:

- **Calibration:** Determines the fixed transformation between sensors (e.g., from camera to robot base).
- **Tracking:** Dynamically computes the pose of moving objects in real-time.

6.1 Point-to-Point Registration (Procrustes Problem)

Used when a set of points with known 1-to-1 correspondences is available (e.g., titanium *fiducial screws* implanted in the bone before the CT scan). Given two matched point sets, A_i (CT) and B_i (Robot), the goal is to minimize the error:

$$\min_{R \in SO(3), t} \sum_{i=1}^N \|A_i - (RB_i + t)\|^2 \quad (6.1)$$

SVD Solution (Closed Form): 1. Calculate the centroids of both point clouds. 2. Center the points by subtracting their respective centroids ($\tilde{A}_i, \tilde{B}_i$). 3. Build the cross-covariance matrix: $H = \sum_i \tilde{B}_i(\tilde{A}_i)^T$. 4. Decompose using SVD: $H = U\Sigma V^T$. 5. The optimal rotation is $\mathbf{R} = VU^T$. 6. If $\det(\mathbf{R}) = -1$ (the algorithm found a reflection instead of a valid rigid rotation), flip the sign of the last column of V and recalculate. 7. The translation is derived from the rotated centroids: $t = \bar{A} - R\bar{B}$.

Iterative Solution (Gauss-Newton): If data is noisy, non-linear least squares optimization is used. The SVD estimate is often used as a highly accurate initial guess (seed) for the Gauss-Newton solver.

6.2 Point-to-Surface Registration (ICP Algorithm)

In less invasive surgeries, screws are not implanted. The surgeon sweeps an optoelectronic probe over the exposed bone surface, acquiring thousands of unknown points. Because it is unknown which specific CT vertex corresponds to each acquired point, the **Iterative Closest Point (ICP)** algorithm is used:

1. Provide an "initial guess" alignment $T^{(0)}$.

2. **Data Association:** For each physically acquired point, geometrically find the closest point on the CT mesh. This creates fictitious pairs.
3. **Resolution:** Apply the SVD Point-to-Point algorithm on these pairs to calculate a new roto-translation $T^{(1)}$.
4. Apply the transformation and loop back to step 2 until the RMS error converges.

ICP guarantees monotonic convergence, but **only to a local minimum**. If the initial guess is poor, ICP might align the front of the real bone with the back of the 3D model. The standard clinical solution is to begin registration by manually touching 3-4 macroscopic anatomical landmarks to provide ICP with an excellent starting condition.

Chapter 7

Bilateral Teleoperation: The Transparency-Stability Dilemma

In **bilateral teleoperation** (e.g., da Vinci 5), the surgeon manipulates a Master console while a Slave robot executes the action on the tissue. The Master sends motion commands to the Slave, and the Slave sends back measured tissue reaction forces to the Master's motors, providing vital **Haptic Feedback**.

This system can be modeled as an electrical circuit using **Two-Port Network theory**, where Effort variables (Voltage $\rightarrow$ Force) and Flow variables (Current $\rightarrow$ Velocity) interact. Using Lawrence's **Hybrid Matrix** $H(s)$, the system is described by connecting inputs to outputs:

$$\begin{bmatrix} f_m \\ -\dot{x}_s \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} \dot{x}_m \\ f_s \end{bmatrix} \quad (7.1)$$

7.1 Transparency (Telepresence)

The clinical goal is to provide the surgeon with the exact physical sensation of touching tissues bare-handed. Let the environment's impedance be $Z_e = f_s/\dot{x}_s$ and the impedance felt at the master be $Z_t = f_m/\dot{x}_m$. **Perfect Transparency** requires $Z_t = Z_e$ regardless of the tissue type. Solving the hybrid matrix equations shows that this identity holds if and only if:

$$H_{11} = 0, \quad H_{22} = 0, \quad H_{12} = 1, \quad H_{21} = -1 \quad (7.2)$$

Physical meaning: In free-space ($f_s = 0$), the console must perfectly cancel its own inertia ($H_{11} = 0$); tissue forces must transfer 1:1 ($H_{12} = 1$); the slave must track master movements infinitely well ($H_{21} = -1$); and if the master is locked, pushing the slave must yield no movement ($H_{22} = 0$).

7.2 Absolute Stability and Llewellyn's Criterion

Teleoperated systems with force feedback close an enormous cyber-physical loop, often plagued by **Time Delays**. In force-feedback systems, communication delays turn control laws into spurious energy generators, causing violent robotic oscillations. This breaks the principle of **Passivity**.

To guarantee clinical safety, the system must possess **Absolute Stability**: it must remain stable regardless of how tightly the surgeon grips the master or what passive tissue the slave hits. For Linear Time-Invariant (LTI) two-port networks, this is verified by **Llewellyn's Absolute Stability Criterion**. Alongside stable poles and individual port dissipativity ($\text{Re}[H_{11}] \geq$

0, $\text{Re}[H_{22}] \geq 0$), Llewellyn's primary constraint requires the Stability Index $\eta(\omega)$ to be ≥ 1 :

$$\eta(\omega) = \frac{2\text{Re}[H_{11}]\text{Re}[H_{22}] - \text{Re}[H_{12}H_{21}]}{|H_{12}H_{21}|} \geq 1 \quad (7.3)$$

7.3 The Inevitable Abyss: Transparency vs. Stability

By plugging the parameters of **Perfect Transparency** (0, 0, 1, -1) into Llewellyn's index, we obtain the most tragic and crucial result of the discipline:

$$\eta(\omega) = \frac{0 - (-1)}{1} = 1 \quad (7.4)$$

A mathematically perfectly transparent surgical teleoperator ***sits exactly on the brink of marginal stability***. In a system where $\eta = 1$, introducing even a single millisecond of network latency or sensor noise causes the index to drop below 1, rendering the machine violently unstable.

This unveils the **Fundamental Trade-off**: *it is physically impossible to achieve maximum tactile transparency and maximum stability simultaneously*. To secure stability ($\eta \gg 1$), engineers must intentionally degrade telepresence: they must inject heavy virtual damping into the joysticks (increasing $\text{Re}[H_{11}]$ and losing the weightless feel) and heavily scale down or filter the force sent back to the surgeon (reducing $|H_{12}H_{21}|$), blurring tactile information in the name of patient safety.

To overcome extreme delays in long-distance telesurgery, modern research uses **Wave Variables**. Instead of exchanging forces and velocities, the ports exchange linear combinations of the two that emulate scattering waves. This mathematical encoding **guarantees network passivity regardless of the time delay**, stabilizing long-distance teleoperation at the cost of slight position tracking degradation.